Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 03
04-08-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Unit Test # 03

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	A	C	D	B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A,B	B,C,D	A,B,C	A,C,D		
SECTION-I (iii)	Q.	11	12	13	14		
	A.	A	D	A	B		
SECTION-II	Q.	1	2	3	4		
	A.	5.00	2.50	1.50	8.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	C	A	C	A	A
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C,D	A,B,D	A,B,C	A,B,C		
SECTION-I (iii)	Q.	11	12	13	14		
	A.	A	C	A	C		
SECTION-II	Q.	1	2	3	4		
	A.	10.00	7.00	278.00	3.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	D	A	A	A	C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C,D	A,B	A,B,C,D	A,B,C		
SECTION-I (iii)	Q.	11	12	13	14		
	A.	B	A	B	A		
SECTION-II	Q.	1	2	3	4		
	A.	3.00	3.00	9.00	2.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**

$$T\sqrt{3} = 6g \Rightarrow T = \frac{6g}{\sqrt{3}} \quad \text{... (i)}$$

Also $T = mg$ ... (ii)

from (i) & (ii)

$$m = \frac{6}{\sqrt{3}} = 2\sqrt{3} = 3.46\text{kg}$$

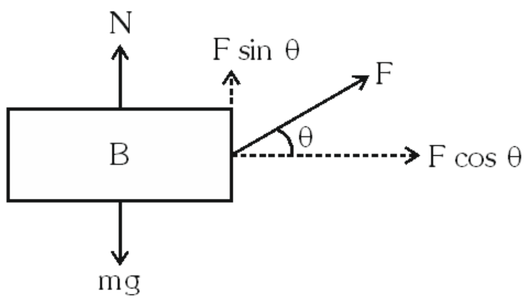
2. **Ans (B)**

In the given situation both the blocks will be at rest. Tension in string is 3N. Thus, static friction acting on block A is 3 N.

4. **Ans (C)**

Let the body is acted upon by a force at an angle q with horizontal.

FBD :



$F\cos\theta = \mu(mg - F\sin\theta)$

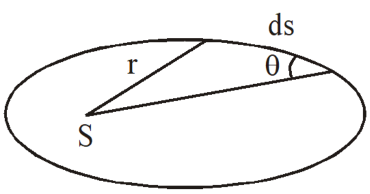
$\Rightarrow F = \dfrac{\mu mg}{\cos\theta + \mu\sin\theta}$. For min. force;

$(\cos\theta + \mu\sin\theta)$ should be max.

$F_{min} = \dfrac{\mu mg}{\sqrt{1+\mu^2}}$

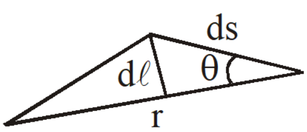
Substituting; $F_{min} = 12.5$ kgf

5. **Ans (D)**



For small displacement ds of the planet its area

can be written as



$dA = \dfrac{1}{2} r d\ell = \dfrac{1}{2} r\, ds \sin\theta$

$A.vel = \dfrac{dA}{dt} = \dfrac{1}{2} r\sin\theta \dfrac{ds}{dt} = \dfrac{Vr\sin\theta}{2}$

$\dfrac{dA}{dt} = \dfrac{1}{2}\dfrac{mVr\sin\theta}{m} = \dfrac{L}{2m}$

6. **Ans (B)**

$Mg' = \dfrac{M}{3} g$

$g' = \dfrac{g}{3}$

$g' = g\left(\dfrac{R}{R+h}\right)^2 = \dfrac{g}{3}$

$\dfrac{R}{R+h} = \dfrac{1}{\sqrt{3}}$.

$h = (\sqrt{3} - 1)R$

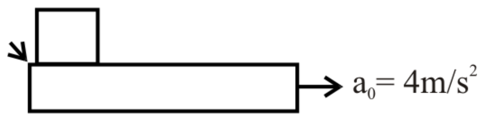
$= (1.732 - 1)6400$

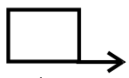
$\boxed{h = 4685\text{km}}$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A,B)**

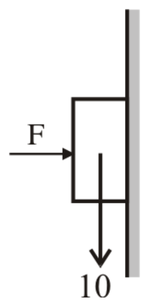




$S = \dfrac{1}{2} \times 2 \times 1 = 1m$ acceleration of block $a_B = \mu g = 2$

$S = \dfrac{1}{2} at^2 = 1m$

8. **Ans (B,C,D)**



$f = \mu N$

$f = \mu N$

$f = \dfrac{1}{2} \times 40$

$f_{max} = 20$ required 10

$F = 30$ N $f = 10$N

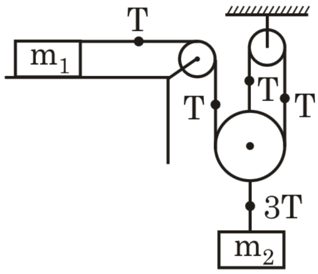
Minimum force $= 10 = \dfrac{1}{2}F$

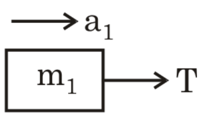
$F = 20$

PART-1 : PHYSICS

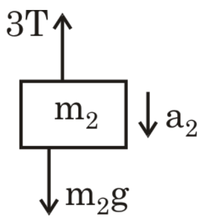
SECTION-I (iii)

11. **Ans (A)**





$T = m_1a_1 \quad ...(i)$



$m_2g - 3T = m_2a_2 \quad ...(ii)$

$a_1 = 3a_2 \quad ...(iii)$

Solve Eq. (i), (ii) and (iii)

$a_1 = 3g/4\ ;\quad a_2 = g/4$

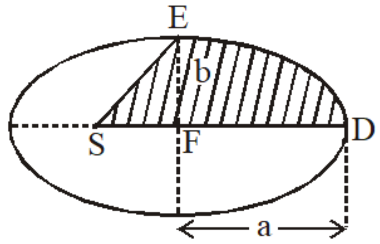
13. **Ans (A)**

$$\frac{A}{T} = \frac{L}{2m} \Rightarrow \frac{L}{mA} = \frac{2}{T}$$

14. **Ans (B)**

$$\frac{dA}{dt} = \text{constant}$$

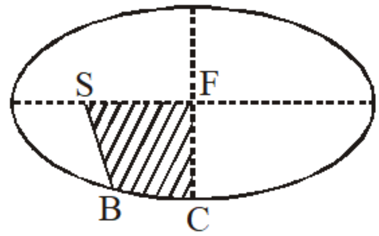
$$\frac{A_1}{t_1} = \frac{A_2}{t_2}$$



$$\frac{A}{40} = \frac{\frac{\pi ab}{4} + \text{Area of } \Delta ESF}{12}$$

$$\frac{A}{40} = \frac{\frac{A}{4} + \text{Area of } \Delta ESF}{12}$$

$$\text{Area of } \Delta ESF = \frac{12}{40}A - \frac{A}{4} = \left(\frac{12-10}{40}\right)A = \frac{A}{20}$$



$$\frac{A}{40} = \frac{\text{Area of SFCB} - \text{Area of } \Delta\text{ SCF}}{1}$$

$$\frac{A}{40} = \text{Area of SFCB} - \text{Area of DESF}$$

$$\text{Area of SFCB} = \frac{A}{40} + \text{Area of } \Delta ESF$$

Total area of shaded region

$$= \text{Area of ESF} + \text{Area of SFCB}$$

$$= \frac{A}{40} + 2\,[\text{Area of ESF}] = \frac{A}{8}$$

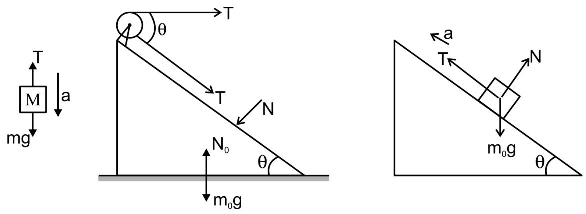
PART-1 : PHYSICS

SECTION-II

1. **Ans (5.00)**

For M : $Mg - T = Ma$ ...(i)

For m : $T - mg\sin\theta = ma$ ...(ii)



and $N = mg\cos\theta$ ...(iii)

For m_0 : $T + T\cos\theta = N\sin\theta$ ...(iv)

[In horizontal direction]

Eliminating a between (i) and (ii),

$$g - \frac{T}{M} = \frac{T}{m} - g\sin\theta$$

$$\Rightarrow T\left(\frac{1}{m} + \frac{1}{M}\right) = g(1+\sin\theta)$$

$$\Rightarrow T = \frac{mMg}{m+M}(1+\sin\theta)$$

Put this value of T and value of N from equation (iii) into equation (iv).

$$\frac{mMg(1+\sin\theta)}{m+M}(1+\cos\theta) = mg\cos\theta \cdot \sin\theta$$

$$\therefore \frac{M}{m+M} = \frac{\cos\theta \cdot \sin\theta}{(1+\sin\theta)(1+\cos\theta)}$$

$$= \frac{\frac{4}{5}\cdot\frac{3}{5}}{\left(1+\frac{3}{5}\right)\left(1+\frac{4}{5}\right)} = \frac{1}{6} \Rightarrow \frac{m}{M} = 5$$

2. **Ans (2.50)**

$2T = \mu \times 10 \times g$

$2T = 0.5 \times 10 \times g$

$2\,Mg = 5\,g$

$M = 2.5$ kg

3. **Ans (1.50)**

From conservation of energy

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{R+h}$$

$$\left(V_0^2 - V^2\right) = 2\frac{GMh}{R(R+h)}$$

$$V_1^2 - V^2 = \frac{4GM}{3R}\ldots\ldots\ldots\ldots.(1)$$

From conservation of angular momentum about centre of earth

$mv_0R = mv(R+h)$

$$v = \frac{v_0R}{R+h} = \frac{v_0}{3}\ldots\ldots.(2)$$

$$\therefore v_0^2 - \frac{v_0^2}{9} = \frac{4}{3}\frac{GM}{R}$$

$$\frac{8}{9}n.\,g.\,R = \frac{4}{3}.\,gr$$

$$n = \frac{3}{2}$$

$n = 1.50$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**

Refer notes.

2. **Ans (C)**

As electropositive character increases basic character of oxide increases.

$$\underbrace{B_2O_3}_{\text{acidic}} < \underbrace{Al_2O_3 < Ga_2O_3}_{\text{amphoteric}} < \underbrace{In_2O_3 < Tl_2O_3}_{\text{basic}}$$

3. **Ans (A)**

$P_1 = 2$ bar, $T = 273$ K, $P_2 = 4$ bar

$$\frac{\Delta U}{w} = \frac{nC_{vm}\Delta T}{\frac{nR\,(T_2 - T_1)}{(x-1)}} = \frac{C_{vm}(x-1)}{R}$$

$$\frac{\frac{3}{2}R(x-1)}{R} = 1.5\,(-1-1) = 1.5 \times (-2) = -3$$

$\frac{P}{V}$ = constant PV^{-1} = constant $x = -1$

4. **Ans (C)**

I Boiling point $\longrightarrow$ intensive

II Entropy $\longrightarrow$ Extensive

III pH $\longrightarrow$ intensive

IV density $\longrightarrow$ intensive

5. **Ans (A)**

$$\Delta S = nC_{p,m} \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

$$\Delta S = 1 \times \frac{5R}{2} \ln 2 + 1 \times R \ln \frac{1}{4}$$

$$\Delta S = \frac{R \ln 2}{2}$$

6. **Ans (A)**

$$\frac{\Delta H_f^o Ag^+(aq)}{\Delta H_f^o Br^-(aq)} = \frac{-5}{6} \quad .... (1)$$

$$-100 = -110 - (-5x + 6x)$$

or, $10 = 5x - 6x \Rightarrow x = -10$

$\therefore \Delta H_f^o,\ Ag^+(aq) = -5x = 50\ KJ\ mol^{-1}$.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C,D)**

$$H_2(g) \longrightarrow 2H(g) = \Delta H_{atomisation}\ H_2(g)$$

$$= 2 \times \Delta H_f H(g)$$

$$= B.EH - H$$

So (C) & (D)

8. **Ans (A,B,D)**

(A) $W = -P \quad dv > 0$

$+ \quad -$

(B) $dH = {}^nC_p dT = 0$

(C) $\Delta S = nR\ln\left(\frac{V_1}{V_2}\right)$; $\Delta S > 0 \because (V_2 < V_1)$

(D) $dG = V\ dP - SdT \quad (dT = 0)$

$+ \quad + \quad +$

9. **Ans (A,B,C)**

Option (A), (B) and (C) are true but option (D) is incorrect because diamond doesn't have Vander Waals force of attraction.

10. **Ans (A,B,C)**

$O \leftarrow \dot{N} = O \longrightarrow$ odd e1 specie

$B.O. = \frac{3}{2} = 1.5$

Due to odd e^- (free e^-) it is paramagnetic.

$NO_2 \Rightarrow 7e^- + 8e^- \times 2 = 23\ e^-$ } not isoelectronic

$CO_2 \Rightarrow 6e^- + 8e^- \times 2 = 22\ e^-$

PART-2 : CHEMISTRY

SECTION-I (iii)

11. **Ans (A)**

$NO^- \longrightarrow$ paramagnetic

O_2^{2-} , $\overline{C}N$, CO $\longrightarrow$ diamagnetic

12. **Ans (C)**

During O_2^- formation , one electron is added to the antibonding molecular orbitals

13. **Ans (A)**

$$\Delta H_3^0 = 3\Delta H_1^0 + 2\Delta H_2^0 = 3(-24) + 2(-34)$$

$$= -240\ kJ/Mol$$

$$\Delta G_3^0 = 3\Delta G_1^0 + 2\Delta G_2^0 = 3(24) + 2(-60)$$

$$= 72 - 120 = -48\ kJ/Mol$$

$$\Delta G_3^0 = \Delta H_3^0 - T\Delta S_3^0 \quad -48 = -240 - 300\ \Delta S_3^0$$

$$\Delta S_3^0 = -640J$$

14. **Ans (C)**

$$T_{eg} = \frac{\Delta H}{\Delta S} = \frac{-24}{-48/300} \Rightarrow T_{eg} = 150\ K$$

$(+24 = -24 - 300\ \Delta S)$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (10.00)**

$\Delta G^0 = \Delta H^0 - T\Delta S$

$\Rightarrow \Delta G^0 = (-54070 - 10 \times 298)$

Also, $\Delta G^0 = (-2.303\ RT \log K)$

$\Rightarrow (-54070 - 10 \times 298)$

$= (-2.303 \times 8.134 \times 298 \log K)$

$\Rightarrow \log K = 10$ Ans: 10

2. **Ans (7.00)**

$w = -1 \times 5 - 10 \times 1 \ln 2 + \frac{1}{2} \times 10 = -10 \ln 2$

$= -10 \times 0.7 = -7\ \text{atm}^{-1}$

3. **Ans (278.00)**

$2C_{(s)} + 3H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow C_2H_5OH_{(l)}$

$(\Delta H_f)_{C_2H_5OH_{(l)}} =$

$\sum (\Delta H_{comb})_{reac\tan t} - \sum (\Delta H_{comb})_{product}$

$= 2 \times (-393.5) + 3(-241.8) - (-1234.7)$

$= -277.7\ \text{kJ/mol}$

4. **Ans (3.00)**

N_3^- linear

NO_2^- bent

I_3^- linear

O_3 bent

SO_2 bent

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$\sin\theta + \operatorname{cosec}\theta = 2 \Rightarrow \sin\theta = 1$

2. **Ans (D)**

$$\cos\frac{\pi}{5}\cos\frac{2\pi}{5}\cos\frac{4\pi}{5}\cos\frac{8\pi}{5} = \frac{\sin\frac{16\pi}{5}}{2^4 \sin\frac{\pi}{5}} = -\frac{1}{16}$$

3. **Ans (A)**

at $x = 5$

$f(x) = 4x + 4$

$\therefore f'(5) = 4$

at $x = 0$ $f(x) = 2x + 2$

$f'(0) = 2$

4. **Ans (A)**

$$4x - 3y - 3x\frac{dy}{dx} + 2y\frac{dy}{dx} + 1 + 2\frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{-4x + 3y - 1}{2y - 3x + 2}$$

5. **Ans (A)**

$\frac{dy}{dx} = 3x^2 = 3t^2$ at 'A'

$\therefore 3t^2 = \frac{T^3 - t^3}{T - t} = T^2 + Tt + t^2$

$T^2 + Tt - 2t^2 = 0$

$(T - t)(T + 2t) = 0 \Rightarrow T = t$ or $T = -2t$

($T = t$ is not possible)

$\frac{m_B}{m_A} = \frac{3T^2}{3t^2} = \frac{4t^2}{t^2}$ (using $T = -2t$)

$m_B = 4$

6. **Ans (C)**

Let point at minimum distance from O is

$(h, h^2 - 4)$

$\therefore OP^2 = h^2 + (h^2 - 4)^2$

$\frac{d(OP^2)}{dh} = 2h + 2(h^2 - 4)2h = 0$

$\Rightarrow h = \pm\sqrt{\frac{7}{2}}, 0$

$\left(\frac{d^2(OP^2)}{dh^2}\right)_{h=\pm\sqrt{\frac{7}{2}}} > 0$

$\therefore$ OP is min at $h = \pm\sqrt{\frac{7}{2}}$

$$OP_{min} = \sqrt{\frac{7}{2} + \left(\frac{7}{2} - 4\right)^2} = \frac{\sqrt{15}}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (C,D)**

$$\frac{1 + \cos(2\alpha + 2\beta) + 1 + \cos(2\alpha - 2\beta)}{2} - \frac{(\cos(2\alpha + 2\beta) + \cos(2\alpha - 2\beta))}{2}$$

$= 1$

8. **Ans (A,B)**

$\cos(x-y) = \frac{1}{2}$ and $\cos(x+y) = \frac{1}{5}$

$\cos x \cos y + \sin x \sin y = \frac{1}{2}$

$\cos x \cos y - \sin x \sin y = \frac{1}{5}$

$\cos x \cos y = \frac{7}{20}, \quad \sin x \sin y = \frac{3}{20}$

$\tan x \tan y = \frac{3}{7}$

$$\frac{\tan(x+y)}{\tan x + \tan y} = \frac{1}{1-\tan x \tan y} = \frac{7}{4}$$

$\cos 2x = \cos(x+y+x-y)$

$\cos 2x = \cos(x+y)\cos(x-y) - \sin(x+y)\sin(x-y)$

$\cos(x+y) = \frac{1}{5}, \ \sin(x+y) = \frac{\sqrt{24}}{5}$

$\cos(x-y) = \frac{1}{2}, \ \sin(x-y) = \frac{\sqrt{3}}{2}$

$$\cos 2x = \frac{1}{10} - \frac{\sqrt{24}}{5}\frac{\sqrt{3}}{2} = \frac{1-\sqrt{72}}{10}$$

9. **Ans (A,B,C,D)**

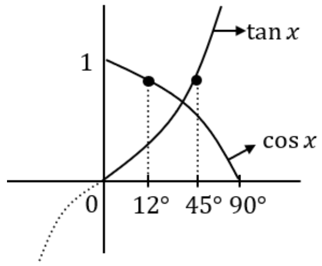
$$\frac{\sin\left(\frac{2\pi}{18^\circ \times 2}\right)\cos\left(\frac{\pi+19\pi}{2\times 18^\circ}\right)}{\sin\frac{2\pi}{180\times x}} = \frac{\sin\frac{\pi}{18}\cos\frac{\pi}{18}}{\sin\frac{\pi}{180}}$$

$\sin\theta + \sin 3\theta + \sin 5\theta + ... + \sin(2n-1)\theta$

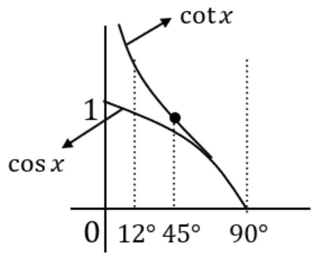
$$= \frac{\sin n\left(\frac{2\theta}{2}\right)\sin\left(\frac{\theta+(2n-1)\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} = \frac{\sin^2 n\theta}{\sin\theta}$$

$$\frac{2\sin\theta}{2\sin\theta}\cos\theta \,.\, \cos 2\theta \,.\, \cos 2^2\theta \,.\, \cos 2^3\theta \,......\, \cos\theta$$

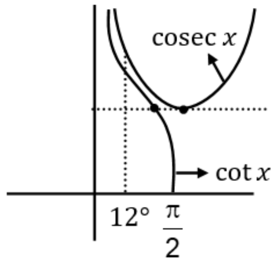
$$= \frac{\sin 2^n\theta}{2^n \sin\theta}$$



clearly $\tan 12^\circ < \cos 12^\circ$



clearly $\cos 12^\circ < \cot 12^\circ$



clearly $\cos 12^\circ < \operatorname{cosec} 12^\circ$

10. **Ans (A,B,C)**

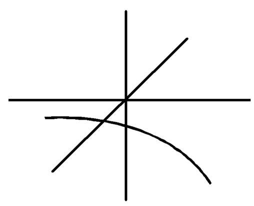
$\tan x = \frac{1}{\sqrt{3}}$ and $\sin x = -\frac{1}{2}$

$x = \pi + \frac{\pi}{6}, \ 3\pi + \frac{\pi}{6}, \ -\pi + \frac{\pi}{6}$

PART-3 : MATHEMATICS

SECTION-I (iii)

11. **Ans (B)**



12. **Ans (A)**

Consider $y = ke^x$ and $y = x$

Let (a, kea) be a point on $y = ke^x$

if it lies on y = x also then $a = ke^\alpha$

$\frac{dy}{dx} = ke^x$

$\therefore \quad \left.\frac{dy}{dx}\right|_{x=\alpha} = ke^\alpha = \alpha = 1 \quad \{\because$ y = x is tangent

to $y = ke^x$ at one point}

$\therefore \quad 1 = ke \qquad$ i.e. k = 1/e

13. **Ans (B)**

$f(x) = -2\sin^2 x + \sin x + 2$

$f(x) = -2\left(\sin^2 x - \frac{\sin x}{2} - 1\right)$

$f(x) = -2\left(\left(\sin x - \frac{1}{4}\right)^2 - \frac{17}{16}\right)$

$= \frac{17}{8} - 2\left(\sin x - \frac{1}{4}\right)^2$

Put $\sin x = \frac{1}{4} \Rightarrow f(x) = \frac{17}{8}$

Put sinx = 1 $\Rightarrow f(x) = \frac{17}{8} - \left(\frac{9}{8}\right)$

Put sinx = – 1 $\Rightarrow f(x) = \frac{17}{8} - 2 \times \frac{25}{8}$

14. **Ans (A)**

$\sum_{r=1}^{10} \sin r\pi + 2\sum_{r=1}^{10} \cos^2 r\pi$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.00)**

$$\frac{\operatorname{cosec}10^\circ + \sec 20^\circ}{\operatorname{cosec}20^\circ + \sec 10^\circ} = \frac{\frac{1}{\sin 10^\circ} + \frac{1}{\cos 20^\circ}}{\frac{1}{\sin 20^\circ} + \frac{1}{\cos 10^\circ}}$$

$$= \frac{\sin 20^\circ \cos 10^\circ}{\sin 10^\circ \cos 20^\circ}\left[\frac{\cos 20^\circ + \sin 10^\circ}{\cos 10^\circ + \sin 20^\circ}\right]$$

$$= \frac{\sin 20^\circ \cos 10^\circ}{\sin 10^\circ \cos 20^\circ}.\left[\frac{\cos 20^\circ + \cos 80^\circ}{\cos 10^\circ + \cos 70^\circ}\right]$$

$$= \frac{\sin 20^\circ \cos 10^\circ}{\sin 10^\circ \cos 20^\circ}\left[\frac{2\cos 50^\circ \cos 30^\circ}{2\cos 40^\circ \cos 30^\circ}\right]$$

$$= \frac{\sin 20^\circ \cos 10^\circ \cos 50^\circ}{\sin 10^\circ \cos 20^\circ \cos 40^\circ}$$

$$= \frac{\cos 10^\circ \cos 50^\circ \cos 70^\circ}{\sin 10^\circ \sin 50^\circ \sin 70^\circ} = \frac{\frac{1}{4}\cos 30^\circ}{\frac{1}{4}\sin 30^\circ}$$

$= \cot 30^\circ = \sqrt{3}$

2. **Ans (3.00)**

$$f'(x) = \frac{1}{1 + \left(\frac{2.3^x}{1+3.9^x}\right)^2}$$

$$2\left\{\frac{(1+3.9^x).3^x\ell n3 - 3^x.(3.9^x\ell n9)}{(1+3.9^x)^2}\right\}$$

put x = –1

$\Rightarrow f'(-1) = \frac{1}{5}\ell n3 \qquad \Rightarrow \qquad e^{5f'(-1)} = 3$

$\tan^{-1}\left(\frac{3.3^x - 3^x}{1 + 3.3^x.3^x}\right) = \tan^{-1}\left(3^{x+1}\right) - \tan^{-1}\left(3^x\right)$

$\Rightarrow f'(x) = \frac{1}{1+(3^{x+1})^2}.3^{x+1}.\ell n3 - \frac{1}{1+(3^x)^2}.3^x.\ell n3$

$\Rightarrow f'(-1) = \frac{1}{5}\ell n3 \qquad \Rightarrow \qquad e^{5f'(-1)} = 3$

3. **Ans (9.00)**

$y^2 = \alpha x^3 - \beta$

$\frac{dy}{dx} = \frac{3\alpha x^2}{2y}$

$\Rightarrow -\frac{dx}{dy}$ at $(2, 3) = \frac{-1}{4}$.

$\alpha = 2,\ \beta = 7$

4. **Ans (2.00)**

$\frac{dy}{dx} = 3ax^2 - 2bx$

$\left.\frac{dy}{dx}\right|_{x=1} = 3a - 2b = 3$

$a = \frac{2b+3}{3} \in [1,2]$

$2b + 3 \in [3,6]$

$2b \in [0,3]$

$b \in \left[0, \frac{3}{2}\right]$